

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T10 · Surds

With Answers

Questions with full worked solutions

■ 28 pages

FAMILY MAP

Surds

F01	Rationalise a surd denominator	9 Q 29 marks
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F02	Simplify and collect surds	2 Q 4 marks
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F03	Multiply and expand surd expressions	2 Q 6 marks
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Working Unit 11

4MA1-1H Q15

ORIGINAL QUESTION

4 marks

15 $(\sqrt{3})^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$k = \dots\dots\dots$
(1)

(b) Show that $\frac{21}{3 - \sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where c and d are integers.

Show each stage of your working clearly.

(3)

Worked Solution 11

Powers and rationalising a surd

UNIT 1
4 marks**Part (a)**

$$(\sqrt{3})^5 = (\sqrt{3})^4 \sqrt{3} = 9\sqrt{3} \Rightarrow k = 9.$$

Use the conjugate

$$\frac{21}{3 - \sqrt{2}} \cdot \frac{3 + \sqrt{2}}{3 + \sqrt{2}} = \frac{21(3 + \sqrt{2})}{9 - 2} = 9 + 3\sqrt{2} = 9 + \sqrt{18}.$$

Final answer

$$k = 9, \quad 9 + \sqrt{18}$$

Question 12

4MA1-1H Q17

ORIGINAL QUESTION

3 marks

- 17 Show that $\frac{1 + \sqrt{5}}{3 - \sqrt{5}}$ can be written in the form $a + \sqrt{b}$ where a and b are integers.

Show each stage of your working clearly.

Worked Solution 12

Rationalise the denominator

MODULAR UNIT 1

3 marks

Conjugate

$$\frac{1 + \sqrt{5}}{3 - \sqrt{5}} \frac{3 + \sqrt{5}}{3 + \sqrt{5}} = \frac{8 + 4\sqrt{5}}{4}.$$

Final answer

$$2 + \sqrt{5}$$

Question 27

4MA1-2H Q15 (b)

ORIGINAL QUESTION UNIT 1

3 marks

- (b) Express $\frac{\sqrt{125} + \sqrt{80}}{\sqrt{3}}$ in the form $\sqrt{n}$ where n is an integer.
Show your working clearly.

.....
(3)

Worked solution 27

Simplify a surd expression

MODULAR UNIT 1

3 marks

(b) – Simplify a surd expression

Simplify the numerator surds

$$\sqrt{125} + \sqrt{80} = 5\sqrt{5} + 4\sqrt{5} = 9\sqrt{5}.$$

Rationalise the denominator

$$\frac{9\sqrt{5}}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3}} = \frac{9\sqrt{15}}{3} = 3\sqrt{15}.$$

Put it in the requested form

$$3\sqrt{15} = \sqrt{9 \cdot 15} = \sqrt{135}.$$

Final answer $\sqrt{135}$

Question 27

4MA1-2H Q23

ORIGINAL QUESTION

4 marks

- 23 The diagram shows a cuboid with a square cross section.

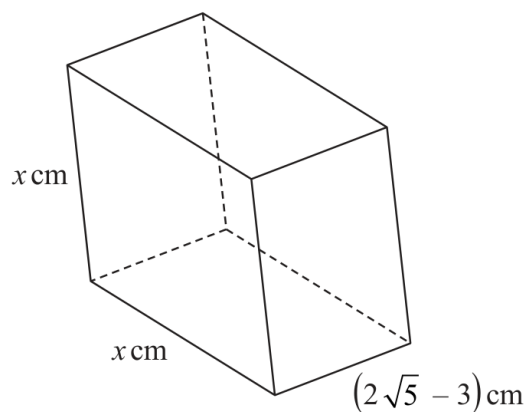


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5}) \text{ cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x = \dots\dots\dots$

Worked Solution 27

Surd volume and a square cross-section

MODULAR UNIT 1

4 marks

Form x squared

$$x^2(2\sqrt{5} - 3) = 13 + 6\sqrt{5} \Rightarrow x^2 = \frac{13 + 6\sqrt{5}}{2\sqrt{5} - 3}.$$

Rationalise

$$x^2 = \frac{(13 + 6\sqrt{5})(2\sqrt{5} + 3)}{11} = 9 + 4\sqrt{5}.$$

Recognise the square

$$9 + 4\sqrt{5} = (2 + \sqrt{5})^2.$$

Take the positive length

$$x = 2 + \sqrt{5}.$$

Final answer

$$2 + \sqrt{5} \text{ cm}$$

Original question 13

4MA1-1H Q17 M01

MODULAR UNIT 1

1 marks

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

.....
(1)

Worked solution 13

Express a surd in the required form

MODULAR UNIT 1

1 marks

M01 – Express a surd in the required form**Factor inside the root**

$$\sqrt{675} = \sqrt{25 \times 27} = 5\sqrt{27}.$$

Final answer

$$n = 5$$

Original question 14

4MA1-1H Q17 M02

MODULAR UNIT 1

3 marks

- (b) Show that $\frac{5 - \sqrt{2}}{\sqrt{2} - 1}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

(3)

Worked solution 14

Rationalise a surd denominator

MODULAR UNIT 1

3 marks

M02 – Rationalise a surd denominator**Multiply by the conjugate**

$$\frac{5 - \sqrt{2}}{\sqrt{2} - 1} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}.$$

Simplify

The denominator is 1 and the numerator is $(5 - \sqrt{2})(\sqrt{2} + 1) = 4 + 4\sqrt{2}$.

Identify integers

$$a = 4, b = 4.$$

Final answer

$$4 + 4\sqrt{2}$$

Original question 11

4MA1-1H Q16 Q16

ORIGINAL QUESTION**3 marks**

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$
Show your working clearly.

Worked solution 11

Chapter: Surds

MODULAR UNIT 1

3 marks

Q16 – Chapter: Surds

Family: Simplify and collect surds • Variant: Add or subtract like surds

Method

$$12/(\sqrt{2} - 1) = 12(\sqrt{2} + 1)/[(\sqrt{2} - 1)(\sqrt{2} + 1)] = 12\sqrt{2} + 12.$$

*Why: Multiply by the conjugate to rationalise the denominator.***Method**

$$(\sqrt{2})^5 = 4\sqrt{2}, \text{ so the left side is } 12\sqrt{2} + 12 - 4\sqrt{2} = 8\sqrt{2} + 12.$$

*Why: Use powers of $\sqrt{2}$ and collect like surds.***Method**

Since $\sqrt{32} = 4\sqrt{2}$, the right side $2\sqrt{32} + 12$ is also $8\sqrt{2} + 12$; therefore the identity is shown.

*Why: Simplify both sides to the same exact surd form.***Final answer****shown identity:** Both sides simplify to $8\sqrt{2} + 12$

Original question 25

4MA1-2H Q17

ORIGINAL QUESTION

3 marks

17 Show that $\frac{\sqrt{12}}{\sqrt{3} + 2}$

can be written in the form $a + \sqrt{b}$ where a and b are integers.

Worked solution 25

Chapter: Surds

MODULAR UNIT 1

3 marks

Q17 – Chapter: Surds

Family: Rationalise a surd denominator • Variant: Binomial surd denominator using a conjugate

Method $\sqrt{12}/(\sqrt{3}+2) \times (\sqrt{3}-2)/(\sqrt{3}-2)$ rationalises the denominator.*Why: Multiply by the conjugate.***Method**The numerator is $6-2\sqrt{12}$ and the denominator is -1 .*Why: Expand and simplify the denominator.***Method**The result is $-6+\sqrt{48}$.*Why: Divide by -1 and write in the required form.***Final answer****surd form:** $-6+\sqrt{48}$

17 (a) Show that $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(3)

Worked solution

January 2020 | Question 17 | 3 marks

Family: Multiply and expand surd expressions • Variant: Expand binomials containing surds

Method / working

1. $\sqrt{12} = 2\sqrt{3}$, so $6 + 2\sqrt{12} = 6 + 4\sqrt{3}$.

2. Square: $(6 + 4\sqrt{3})^2 = 84 + 48\sqrt{3}$.

3. Factor 12: $84 + 48\sqrt{3} = 12(7 + 4\sqrt{3})$.

Final answer: $12(7 + 4\sqrt{3})$

20 The area of a rectangle is 18 cm^2

The length of the rectangle is $(\sqrt{7} + 1)\text{ cm}$.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.

Give your answer in the form $a\sqrt{b} + c$ where a , b and c are integers.

..... cm

Worked solution

January 2020 | Question 20 | 3 marks

Family: Rationalise a surd denominator • Variant: Binomial surd denominator using a conjugate

Method / working

1. $\text{Width} = 18/(\sqrt{7}+1)$.
 2. Multiply by the conjugate $(\sqrt{7}-1)/(\sqrt{7}-1)$.
 3. $18(\sqrt{7}-1)/(7-1) = 3\sqrt{7}-3$ cm.
- Final answer: $3\sqrt{7}-3$ cm

- 18** Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is a prime number and y is an integer, find the value of x and the value of y . Show each stage of your working clearly.

$x =$

$y =$

Worked solution

January 2021 | Question 18 | 3 marks

Family: Multiply and expand surd expressions • Variant: Expand binomials containing surds

Method / working

1. Expand $(8-\sqrt{x})(5+\sqrt{x})=40+3\sqrt{x}-x$.
2. Compare with $y\sqrt{x}+21$: $x-19=0$, so $x=19$.
3. Then $y=3$.

Final answer: $x=19$, $y=3$

17 Express $\frac{8}{\sqrt{5}-1}$ in the form $\sqrt{a} + b$ where a and b are integers.

Show each stage of your working clearly.

Worked solution

November 2021 | Question 17 | 3 marks

Family: Rationalise a surd denominator | Variant: Binomial surd denominator using a conjugate

METHOD / WORKING

1. $\frac{8}{\sqrt{5}-1}$

2. Rationalise the denominator:

3. $\frac{8}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1}$

4. $= \frac{8(\sqrt{5}+1)}{5-1}$

5. $= 2(\sqrt{5}+1) = 2\sqrt{5}+2$

6. In the form $\sqrt{a}+b$,

7. $2\sqrt{5}+2 = \sqrt{20}+2$

8. Answer: $\sqrt{20}+2$

FINAL ANSWER

$\sqrt{20}+2$

19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

(Total for Question 19 is 3 marks)

Answer reference | May-Nov 2020 | Question 19

Family: Rationalise a surd denominator • Variant: Single-term surd denominator

19	$\frac{6}{3-\sqrt{7}} \times \frac{3+\sqrt{7}}{3+\sqrt{7}}$ or $\frac{6}{3-\sqrt{7}} \times \frac{-3-\sqrt{7}}{-3-\sqrt{7}}$			M1
	$\frac{6(3+\sqrt{7})}{3^2-7}$ or $\frac{6(3+\sqrt{7})}{2}$ or $\frac{6(-3-\sqrt{7})}{-3^2+7}$ or $\frac{6(-3-\sqrt{7})}{-2}$			M1 (numerator may be expanded or denominator may be 4 terms which need to be all correct)
		$9+3\sqrt{7}$	3	A1 dep on M2 for $9+3\sqrt{7}$ or $3(3+\sqrt{7})$ from correct working
				Total 3 marks

Worked solution

May-Nov 2020 | Question 19 | 3 marks

Family: Rationalise a surd denominator • Variant: Single-term surd denominator

Method

$$(6)/(3-\sqrt{7})$$

Rationalise the denominator:

$$(6)/(3-\sqrt{7}) \times (3+\sqrt{7})/(3+\sqrt{7})$$

$$= (6(3+\sqrt{7}))/((9-7))$$

$$= (6(3+\sqrt{7}))/2 = 3(3+\sqrt{7})$$

$$= 9+3\sqrt{7}$$

Answer: $9+3\sqrt{7}$